VAPT Lab Report

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1. Executive Summary

In this lab, we performed a Vulnerability Assessment and Penetration Testing (VAPT) on a test system, **Metasploitable 2 (192.168.56.102)**, using **Kali Linux** as the attacker machine. The goal was to:

- Identify system and network vulnerabilities using **open-source tools**.
- Analyze risks based on **CVSS scoring** and likelihood/impact.
- Document findings and propose **remediation strategies**.

The assessment included **network scanning (Nmap)**, **web server testing (Nikto)**, and **risk evaluation**. Screenshots were taken at each step for reference. This report summarizes the methodology, findings, and recommendations in a structured manner.

OpenVAS (Greenbone Vulnerability Management)

OpenVAS is an open-source vulnerability scanning tool that is part of the Greenbone Vulnerability Management (GVM) framework. It is used to identify security weaknesses in systems, servers, and network devices.

Unlike simple port scanners, OpenVAS performs deep vulnerability analysis by checking services against known security databases such as CVE (Common Vulnerabilities and Exposures).

2. Objective of the Task

- Understand **security assessment and testing** techniques.
- Follow a **VAPT methodology** from planning to reporting.
- Learn to align testing with **security standards and compliance** (e.g., ISO 27001, GDPR).
- Identify **common vulnerabilities** in network and web services.
- Practice **risk assessment** and document results in an organized report.

3. Scope of the Lab

- **Target System:** Metasploitable 2 VM (Intentionally vulnerable)
- **Attacker System:** Kali Linux VM
- **Environment:** VirtualBox host-only network, ensuring isolation from the host internet
- **Purpose:**
  - Practice network and web vulnerability scanning
  - Identify weaknesses without using paid tools
  - Understand risk assessment basics and proper documentation

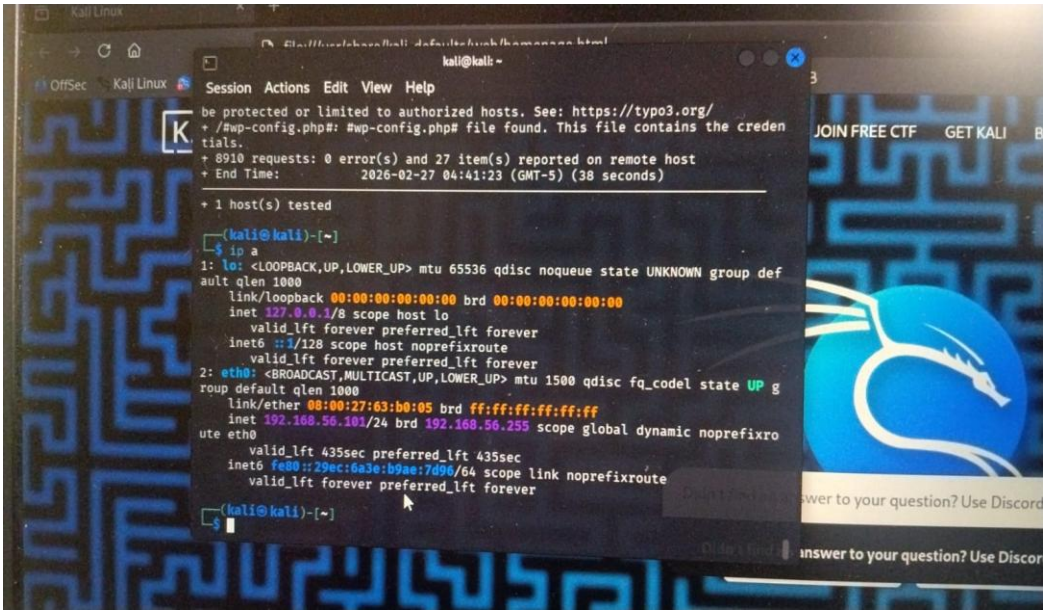

Fig: ***Attacker System IP***

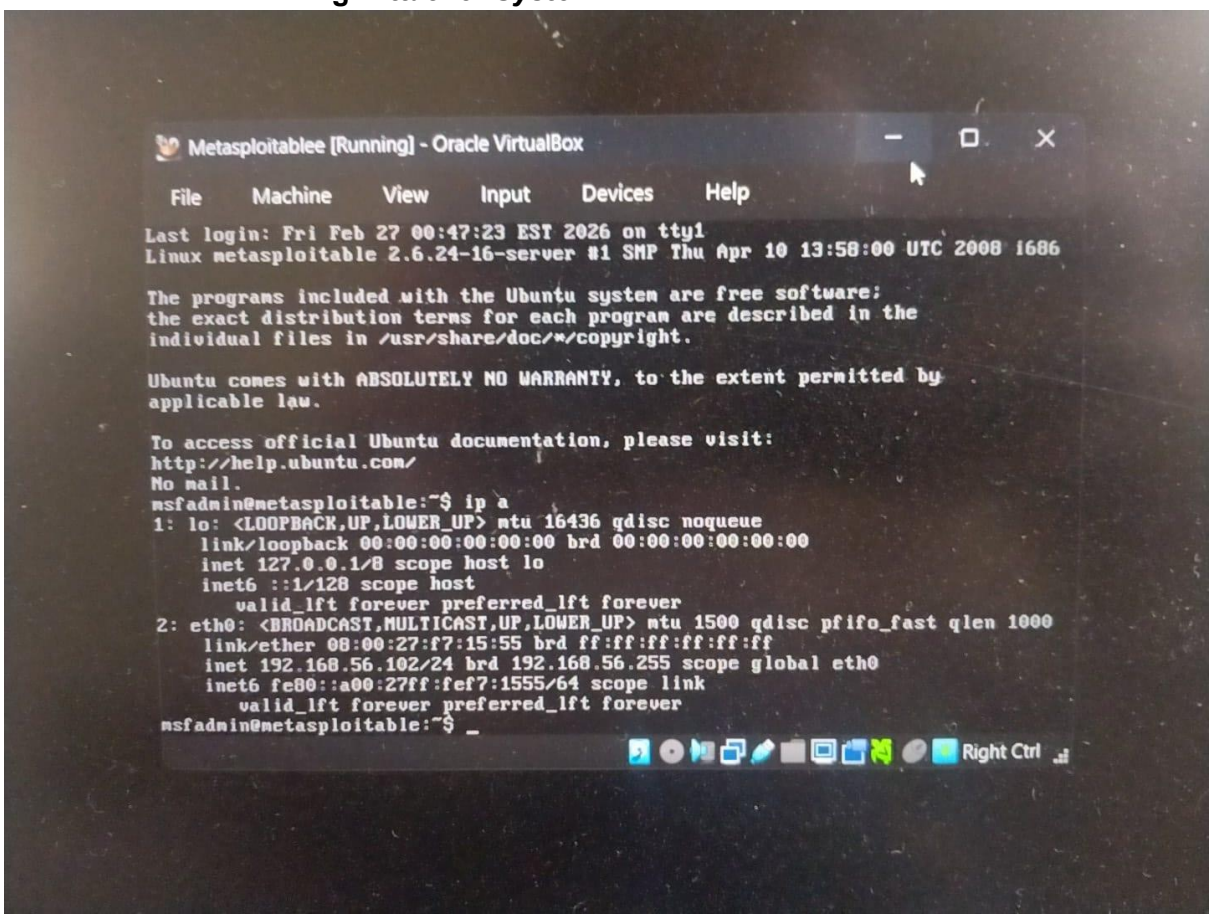

Fig*: Target System Ip*

3. Methodology

The following **structured VAPT approach** was followed:

Step 1 – Planning

- Defined target IP: 192.168.56.102
- Ensured testing is performed in a safe, isolated lab environment

Step 2 – Discovery

- **Network scanning with Nmap**:
  - Command used: nmap -A 192.168.56.102
  - Purpose: Identify open ports, services, and versions

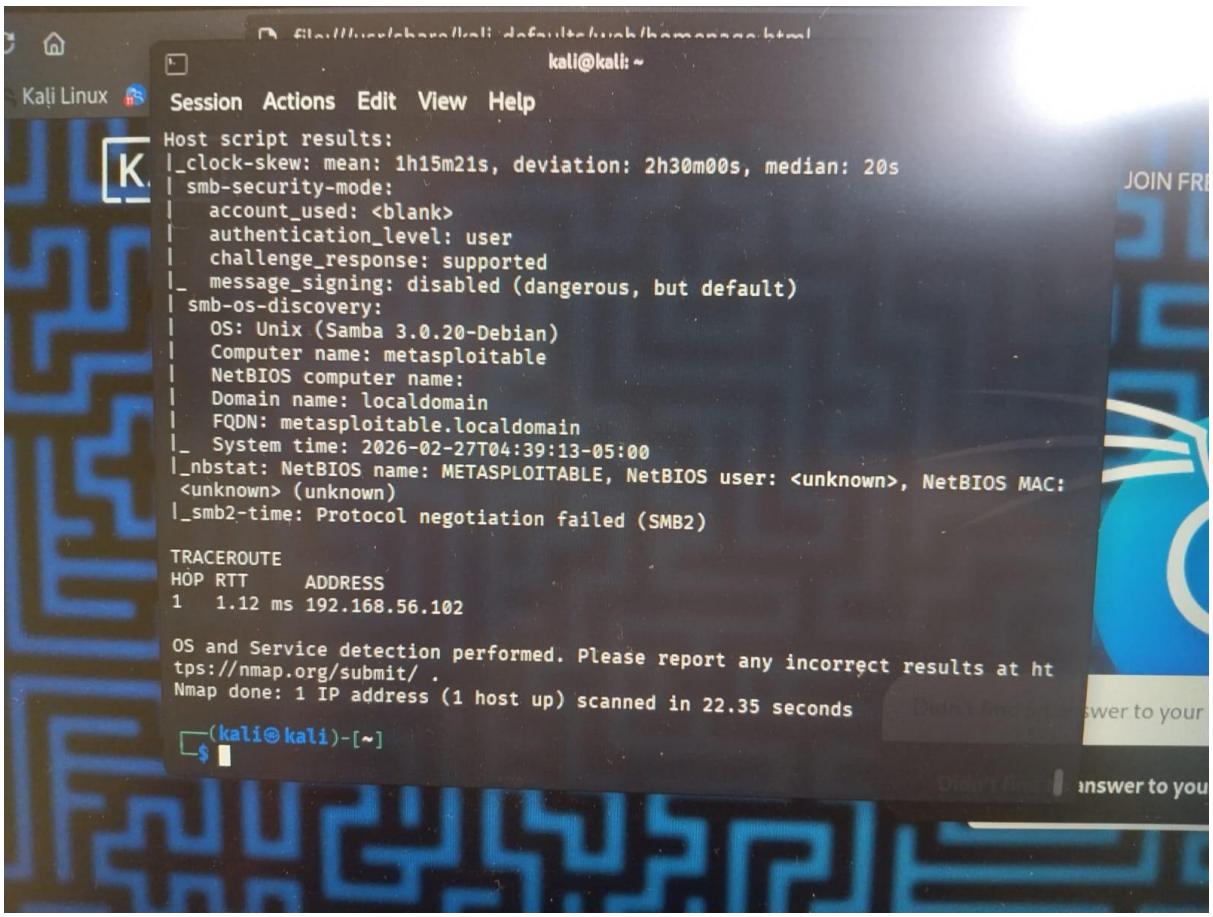

Fig: *Open ports and service versions detected on Metasploitable 2*

- **Web server check**:
  - Opened target in browser: http://192.168.56.102
  - Verified Apache web server presence

Step 3 – Vulnerability Testing

- **Nikto scan**:
  - Command: nikto -h http://192.168.56.102
  - Purpose: Detect outdated web server versions, misconfigurations, and common vulnerabilities

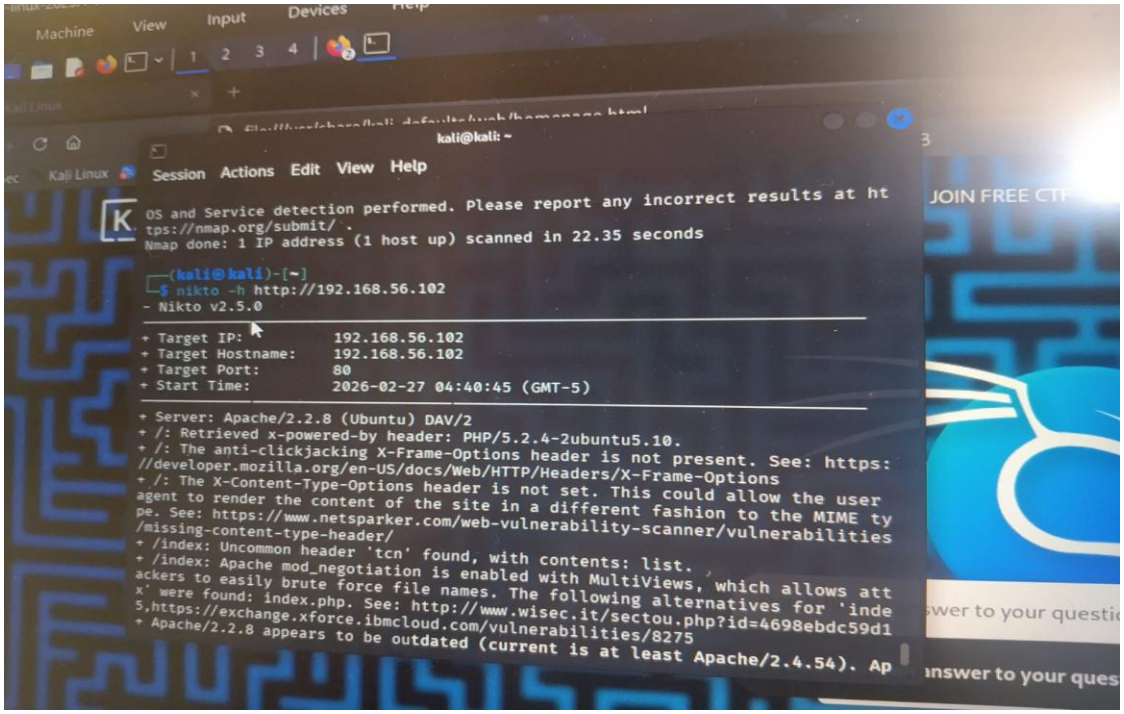

Fig: *Web server vulnerabilities detected using Nikto*

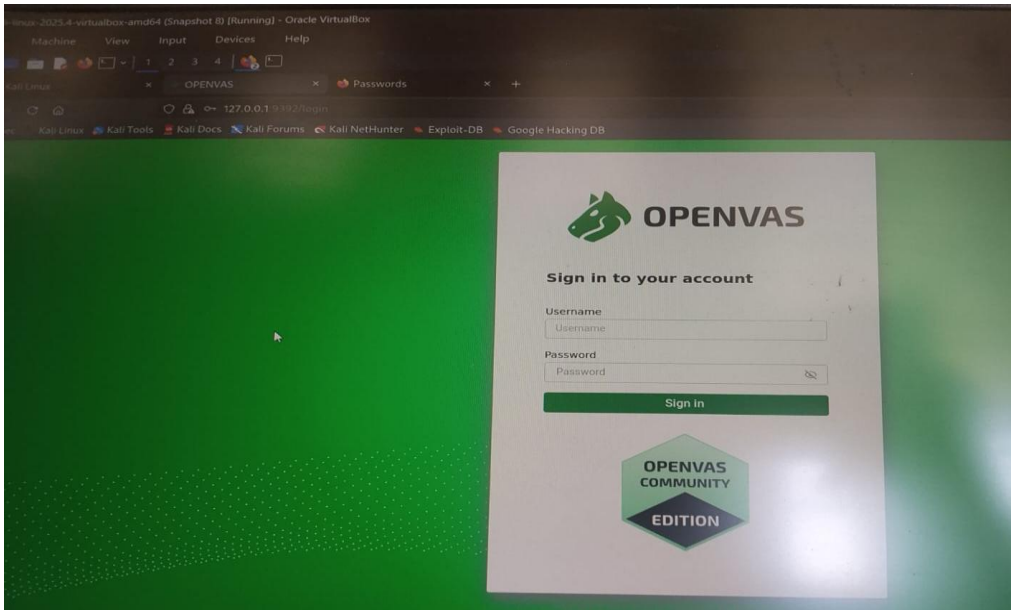

Fig: *Greenbone Security Assistant (OpenVAS Web Interface Login Page)*

Risk Analysis:
- CVSS score applied to each vulnerability
- Risk levels assigned (High, Medium, Low)
- Risk matrix created in Excel

Step 4 – Reporting
- Screenshots recorded for all steps
- Findings organized in tables for clarity
- Recommendations provided for each identified vulnerability

4. Tools Used

Tool	Purpose
Kali Linux	Attacker machine
Nmap	Network discovery and service enumeration
Nikto	Web server vulnerability scanning
Metasploitable 2	Target machine for testing
Excel / Google Sheets	Risk assessment and risk matrix
Firefox	Web testing and screenshot capture

5. Technical Findings

5.1 Network Scan (Nmap)

Port	Service	Version	Vulnerability	Risk Level
21	FTP	vsftpd 2.3.4	Known backdoor	Critical
22	SSH	OpenSSH 4.7p1	Weak version	Medium
23	Telnet	telnetd	Unencrypted login	High
80	HTTP	Apache 2.2.8	Outdated server	High
445	SMB	Samba 3.0.20	Vulnerable version	High

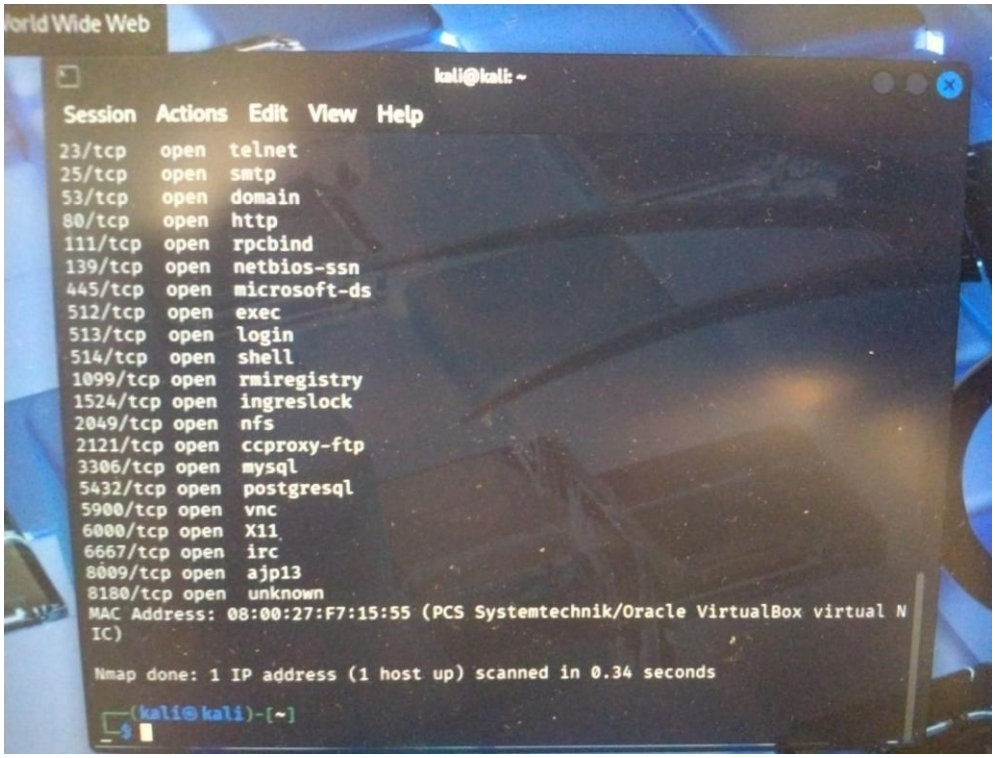

Fig: *Nmap Scan Results Showing Open Ports and Running Services*

5.2 Web Vulnerability Scan (Nikto)

Finding	Description	Risk Level
Outdated Apache	Apache 2.2.8 version	High
Directory indexing enabled	Exposes file structure	Medium
TRACE method enabled	Can be exploited	Medium
Sensitive files	phpinfo() accessible	High

5.3 Risk Matrix

Likelihood / Impact	Low	Medium	High
Low			
Medium		Directory indexing	
High	TRACE method		FTP backdoor, Telnet, Apache outdated

Risk Matrix – Metasploitable 2 Vulnerabilities

Likelihood / Impact	Low	Medium	High
Low			
Medium		Directory indexing	
High	TRACE method		FTP backdoor, Telnet, Apache outdated

Fig: *Prioritization of vulnerabilities based on likelihood and impact*

6. Recommendations

1. **FTP Service (vsftpd 2.3.4)** → Upgrade to latest secure version or disable if not needed.
2. **Telnet Service** → Disable Telnet, use SSH with strong password.
3. **Apache Web Server** → Update to latest version, remove unnecessary modules.
4. **Directory Listing** → Disable directory indexing in Apache config.
5. **Sensitive Files** → Remove unnecessary files, restrict access.
6. **SMB / Samba** → Patch or disable vulnerable services.

All recommendations aim to **reduce attack surface** and improve lab system security.

7. Conclusion

The lab exercise demonstrated:

- How to perform a **complete vulnerability assessment** using free tools.
- The importance of **network scanning** (Nmap) and **web scanning** (Nikto) to detect weaknesses.
- How to **categorize vulnerabilities**, assign **risk levels**, and document findings for reporting.

Performing this exercise helps build **practical skills in ethical hacking, risk assessment, and reporting** in a controlled environment.

- **Attachments (Screenshots)**

1. Nmap scan results
2. Nikto scan results
3. Web server page screenshot
4. Risk matrix Excel screenshot